Let c_r be the multiplicity of r in S, and associate to S the polynomial

$$P_S(x) = \sum_{r=1}^{19} c_r x^r.$$

If $S = A_i \sqcup B_i$ and $A_i = i + B_i$, then, writing $Q_i(x)$ for the multiplicity polynomial of B_i,

$$P_S(x) = (1 + x^i)Q_i(x),$$

where Q_i has nonnegative integer coefficients. Conversely, any such factorisation gives the required partition. Thus the condition for i is equivalent to

$$\frac{P_S(x)}{1 + x^i} \in \mathbb{Z}_{\geq 0}[x].$$

0.0.1 1. Determining the maximum index

The relevant factorizations are

$$1 + x^3 = (1 + x)(1 - x + x^2), \; 1 + x^5 \; = (1 + x)(1 - x + x^2 - x^3 + x^4), \; 1 + x^6 = (1 + x^2)(1 - x^2 + x^4).$$

The displayed factors are the distinct cyclotomic polynomials of orders $2, 4, 6, 8, 10, 12$, so they are pairwise coprime. Hence

$$L_5 = \operatorname{lcm}_{1\leq i\leq 5}(1 + x^i) \;=\; (1 + x)(1 + x^2)(1 - x + x^2)(1 + x^4)(1 - x + x^2 - x^3 + x^4),$$

and

$$L_6 = L_5(1 - x^2 + x^4).$$

Their degrees are

$$\deg L_5 = 13, \qquad \deg L_6 = 17.$$

Both products are palindromic. From their factors,

$$[x]L_5 = [x^{12}]L_5 = -1, \qquad [x]L_6 = [x^{16}]L_6 = -1,$$

and both have constant and leading coefficient 1.

Suppose some S had index at least 6. Then $L_6 \mid P_S$. Since $\deg P_S \leq 19$ and $P_S(0) = 0$, we may write

$$P_S(x) = (ax + bx^2)L_6(x)$$

for integers a, b. Using 4, three coefficients of P_S are

$$[x]P_S = a, \qquad [x^2]P_S = b - a, \qquad [x^{18}]P_S = a - b.$$

They must all be nonnegative, so

$$b \geq a, \qquad a \geq b.$$

Thus $a = b > 0$, since S is nonempty. Therefore

$$\frac{P_S(x)}{1 + x} = axL_6(x).$$

But the coefficient of x^2 in this quotient is

$$a[x]L_6 = -a < 0,$$

contradicting 1. Hence no multiset has index at least 6, so

$$N \leq 5.$$

0.0.2 2. Determining the least possible maximum element

Suppose S satisfies the conditions for $i = 1, \ldots, 5$ and has maximum element at most 15. Then $L_5 \mid P_S$, and hence

$$P_S(x) = (ax + bx^2)L_5(x).$$

As before,

$$[x]P_S = a, \qquad [x^2]P_S = b - a, \qquad [x^{14}]P_S = a - b.$$

Nonnegativity again gives $a = b > 0$. Consequently,

$$\frac{P_S(x)}{1+x} = axL_5(x),$$

whose x^2-coefficient is $-a$, impossible. Thus any multiset satisfying the first five conditions has maximum element at least 16.

Now consider

$$P_0(x) = x\prod_{j=1}^{5}(1+x^j).$$

Using the factorizations above,

$$\prod_{j=1}^{5}(1+x^j) = (1+x)^2 L_5(x).$$

The polynomial P_0 has nonnegative integer coefficients and degree

$$1 + 1 + 2 + 3 + 4 + 5 = 16.$$

Moreover, for every $1 \le i \le 5$,

$$\frac{P_0(x)}{1+x^i} = x \prod_{1 \le j \le 5 \ne i} (1+x^j)$$

has nonnegative integer coefficients. Therefore its corresponding multiset has index at least 5, and hence exactly 5 by 5. Its maximum element is 16. Thus

$$N = 5, \qquad M = 16.$$

0.0.3 3. Minimising the number of elements

Let S now have index 5 and maximum element 16. Since $L_5 \mid P_S$,

$$P_S(x) = (ax + bx^2 + cx^3)L_5(x)$$

for integers a, b, c.

Put

$$D_1(x) = \frac{L_5(x)}{1+x}.$$

Because

$$L_5 = 1 - x + \cdots - x^{12} + x^{13},$$

comparison in $L_5 = (1+x)D_1$ gives

$$D_1(x) = 1 - 2x + \cdots - 2x^{11} + x^{12}.$$

By 1, the polynomial

$$\frac{P_S}{1+x} = (ax + bx^2 + cx^3)D_1$$

has nonnegative coefficients. Its coefficients of x, x^2, x^{14}, x^{15} are respectively

$$a, \qquad b - 2a, \qquad b - 2c, \qquad c.$$

Hence

$$a, c \geq 0, \qquad b \geq 2a, \qquad b \geq 2c.$$

Next put

$$D_2(x) = \frac{L_5(x)}{1+x^2}.$$

The beginning of L_5 is

$$L_5(x) = 1 - x + 2x^2 + \cdots,$$

so comparison in $L_5 = (1+x^2)D_2$ gives

$$D_2(x) = 1 - x + x^2 + \cdots.$$

The coefficient of x^3 in

$$\frac{P_S}{1+x^2} = (ax + bx^2 + cx^3)D_2$$

is $a - b + c$. Therefore

$$b \leq a + c.$$

Combining 9 and 10,

$$2a \leq b \leq a + c \quad \Longrightarrow \quad a \leq c,$$

while

$$2c \leq b \leq a + c \quad \Longrightarrow \quad c \leq a.$$

Thus $a = c$, and then necessarily

$$b = 2a.$$

Since S is nonempty, a is a positive integer. Equation 8 becomes

$$P_S(x) = a, x(1+x)^2 L_5(x) = a, x \prod_{j=1}^{5}(1 + x^j).$$

The number of elements in S, counted with multiplicity, is

$$|S| = P_S(1) = a \prod_{j=1}^{5}(1+1) = 32a \geq 32.$$

The construction 6 has $a = 1$, so equality is attained.

$$\boxed{|S|_{\min} = 32}$$